Swiss Federal Institute of Technology Zurich

Seminar for Statistics

Department of Mathematics

Bachelor Thesis Spring 2026

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On Why Scoring Functions Cannot Assess Tail Properties: A More Detailed Analysis

Submission Date: 14 June 2026

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Acknowledgements

A warm thank you goes out to everybody who supported me in writing this thesis.

Firstly, thank you to my supervisors Prof. Ziegel and Dr. Pic, who gave advice on all parts of the process.

Furthermore, I would like to thank Dr. Alban Moor, whose comments on my intuition led to the examples in Section 3.3.

Finally, my friends Philipp, Roya and Martin deserve a big thank you for proofreading this thesis.

Code and data availability

The R code for the simulations and plots in Chapter 5 can be found in a single R markdown file under `https://github.com/jostlu/bachelor_thesis`.

Abstract

This thesis explores different areas of forecast evaluation focusing on vector and real valued point forecasts and probabilistic forecasts. We show limitations in assessment of statistical functionals that give us information about the tail of a distribution. These functionals come into play if the extreme behaviour of a distribution is of interest rather than the average behaviour. To this end, we introduce max-functionals, analyse their role in tail behaviour and see that the aforementioned limitations arise.

The problems we see still cannot be avoided even if the whole distribution is reported as all scoring rules cannot reliably distinguish between functionals of different max-functional value. We show this problem from a theoretic point of view and with a simulation. Combined with further problems that come up in practice, one needs to be careful when dealing with max-functionals.

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Notation

The following notation will be used. The symbols and definitions are in line with Brehmer and Strokorb [2019], although we introduce some additional ones. Note that this table is intended for looking up definitions and the symbols will be defined throughout the thesis.

Symbol	In	Meaning
O	2.1	Observation domain, i.e. domain of possible observations
$\mathcal{O}$	2.1	The Borel σ-algebra of O
$\mathcal{F}^*$	2.1	The set of all probability distributions on $(\mathrm{O}, \mathcal{O})$
$\mathcal{F}$	2.1	A convex set of probability distributions, $\mathcal{F} \subseteq \mathcal{F}^*$
A	2.1	Action domain, i.e. domain of possible forecasts
T	2.1	A functional, $T : \mathcal{F} \to \mathrm{A}$
S	2.2	A scoring function, $S : \mathrm{A} \times \mathrm{O} \to \mathbb{R}$
$T_k(F)$	2.3	The k-th moment of the distribution F
$U_{[a,b]}$	2.2	The uniform distribution on $[a, b] \subseteq \mathbb{R}$
$\mathcal{F}_{\text{unif}}$	2.2	The family of uniform distribution on $\mathbb{R}$
$\mathcal{E}_k(\mathcal{F})$	2.4	The set of $\mathbb{R}^k$-valued elicitable functionals defined on $\mathcal{F}$
$\mathrm{elic}(T)$	2.4	The elicitation complexity of T w.r.t. some class $(\mathcal{C}_k)_{k\in\mathbb{N}}$
$\overline{F}(x)$	3	The survival function of a distribution F, $\overline{F}(x) = 1 - F(x)$
x^F	3.2	The upper endpoint of F
$\mathcal{F}_{<\infty}$	3.2	Bounded from above probability distributions
x_F	3.3	The lower endpoint of F
$\mathbb{1}(A)$	3.3	The characteristic/indicator function on the set A
$F_{[\alpha,\beta]}$	3.3	The α, β-quantilised distribution of F
$\mathbb{S}$	4.1	A scoring rule, $\mathbb{S} : \mathcal{F} \times \mathrm{O} \to \mathbb{R}$
E_θ	5.1	The exponential distribution with rate $\theta \in (0, \infty)$
$\mathcal{F}_{\text{exp}}$	5.1	Family of convex combinations of exponential distributions
T_ρ	5.1	Tail index of a distribution in $\mathcal{F}_{\text{exp}}$
CRPS	5.2	Continuous ranked probability score

Chapter 1

Introduction

This thesis gives an introduction to the concepts and key findings of scoring functions and elicitability, building on the paper "Why scoring functions cannot assess tail properties" by Brehmer and Strokorb [2019].

Whether it is tomorrow's weather, the amount of people that take the same train as yourself in the morning or the price of a stock, most future events cannot be foretold with certainty. We usually model these quantities as a random variable Y which follows an unknown distribution F and frequently, one needs to make decisions based on F. Such decisions include whether or not to bring an umbrella or how much money to invest. In important situations, one needs to make sure this decision is based on well estimated properties of F. Common properties are for example the mean, the variance or quantiles.

Thus, we need to estimate these properties and more importantly, we need to know which estimates are good. That's the point where forecast evaluation comes into play. For point forecasts, we can rank competing forecasts using a scoring or loss function S. Given a forecast x and an observation y, S determines how good the forecast was. For some properties, it is clear which loss function we need to choose, i.e. for the mean it is $S(x, y) = (y - x)^2$ as the mean is the minimiser of $\mathbb{E}_F[(Y - t)^2]$. For several properties, there is no such scoring function, which makes the assessment of the forecast harder. In some cases, the problem can be avoided by considering more properties of the distribution jointly but we show that this is not always the case.

We begin by introducing a framework to assess point forecasts in Chapter 2. Chapter 3 then examines max-functionals, which arise when thinking about the tail behaviour of distributions. Finally, we turn our back on point forecasts as the previous chapters indicate that point forecasts come with some drawbacks. Chapter 4 treats scoring rules, which are at the heart of probabilistic forecasting, where the whole estimated distribution of Y, and not just some properties of it, is assessed based on observations. We find that there still are problems and conclude with a simulation study in Chapter 5 showing these weaknesses not only theoretically but also with issues that could arise in applications.

Chapter 2

Theoretical Background

In this chapter, we examine scoring functions and elicitability, as they are of central interest in forecast evaluation. We consider a random process that gives us (real- or vector-valued) data and say that they are realisations of a random vector (including the one dimensional case) Y. Y follows the distribution F which we do not know. We first look at point forecasts, which make predictions about certain properties of F and try to evaluate which point forecast is the best.

2.1 The setting

Let us first introduce the needed notation. We denote by $\mathrm{O} \subseteq \mathbb{R}^d$ the observation domain and by $\mathcal{O}$ its Borel σ-algebra coming from the induced euclidean topology of $\mathbb{R}^d$. $\mathcal{F}^*$ is then the set of all probability distributions on $(\mathrm{O}, \mathcal{O})$, which means it contains the possible distributions of Y. We do usually not consider all distributions as we make assumptions about F like that it has finite second moment or that it belongs to a parametrised family of distributions. This is why we look at non-empty subsets $\mathcal{F} \subseteq \mathcal{F}^*$ for which certain conditions hold.

$\mathrm{A} \subseteq \mathbb{R}^n$ is our action domain, which means that it is the domain where our possible point forecasts lie. A does not need to be the same as O and does not even have to have the same dimension as O because the realisations of Y lie in O and we are not trying to predict the outcome of another observation of Y but forecast some property of Y's distribution via a functional.

A function $T : \mathcal{F} \to \mathrm{A}$ is called functional, it tells us something about a probability distribution. The most common example is the mean of a distribution (if it exists), in which case it holds that $T_1(F) = \int_{\mathrm{O}} y dF(y)$. Similarly, a function which maps a distribution to one of its moments (again, if it exists) and the variance are functionals. Of interest may also be some quantile of a distribution, the mapping to which is a functional too. In the rest of this thesis, we say that the mean is a functional when the formally correct expression would be that the mapping of a distribution to its mean is a functional. We only consider vector valued functionals but it is possible to extend the definitions to e.g. set valued functionals, an example of which would be the mapping to some level α confidence set.

Definition 2.1. *A measurable function $h : O \to \mathbb{R}$ is called $\mathcal{F}$-integrable if it is integrable with respect to all $F \in \mathcal{F}$. Similarly, $S : A \times O \to \mathbb{R}$ is called $\mathcal{F}$-integrable if $y \mapsto S(x, y)$ is $\mathcal{F}$-integrable $\forall x \in A \subseteq \mathbb{R}^n, F \in \mathcal{F}$.*

For these functions, we use the following notation:

$$\overline{h}(F) := \int_{\mathrm{O}} h(y)dF(y) \text{ and } \overline{S}(x,F) := \int_{\mathrm{O}} S(x,y)dF(y).$$

This bar notation is just an expectation with respect to the given probability distribution, i.e. $\overline{h}(F) = \mathbb{E}_F[h(Y)]$ and $\overline{S}(x,F) = \mathbb{E}_F[S(x,Y)]$. Note that the definition of $\mathcal{F}$-integrability is an equivalent way of saying that those expectations exist.

2.2 Scoring functions and consistency

Definition 2.2. *$S : A \times O \to \mathbb{R}$ is called a scoring function if it is $\mathcal{F}$ integrable. A scoring function is called $\mathcal{F}$-consistent for a functional $T : \mathcal{F} \to A$ if $\overline{S}(x,F) \geq \overline{S}(T(F),F)$ holds for all $x \in A$ and $F \in \mathcal{F}$. S is strictly consistent for T if, in addition, the equality $\overline{S}(x,F) = \overline{S}(T(F),F)$ implies $T(F) = x$ for all $x \in A$ and $F \in \mathcal{F}$.*

From now on, we usually omit the $\mathcal{F}$ in $\mathcal{F}$-consistent if it is clear from the context.

In essence, scoring functions return an assessment of how good our prediction x of the property F was compared to a realisation y of Y. We use the convention that a lower score means a better prediction. This might be misleading since a score alone does not tell us anything. Take, for example, $S(x,y) = 0\ \forall x,y$. This scoring function always returns a low score but is obviously unusable for comparing different forecasts.

An equivalent way of interpreting a scoring function is as a loss function. In this setting, we choose S to penalize bad forecasts. Of course, "bad" depends on the application and is up to our own discretion. This is discussed further in Section 2.5.

Example 2.3. To illustrate this definition, take $\mathrm{O} = \mathrm{A} = \mathbb{R}$, define $U_{[a,b]}$ to be the uniform distribution on the interval $[a,b]$ for $a < b$ and take $\mathcal{F}_{\text{unif}} := \{U_{[a,b]} \mid a < b \in \mathbb{R}\}$. We then have that $S_1(x,y) := (x-y)^2$ is consistent with the mean $T_1(U_{[a,b]}) = \frac{a+b}{2}$ as

$$\begin{aligned}
\overline{S}_1(x,U_{[a,b]}) &= \int_{\mathbb{R}} (x-y)^2 dU_{[a,b]}(y) \\
&= \int_{[a,b]} (x-y)^2 \frac{dy}{b-a} \\
&= \frac{1}{b-a} \int_a^b (x^2 - 2xy + y^2) dy \\
&= \frac{1}{b-a} (x^2(b-a) - 2x\frac{b^2-a^2}{2} + \frac{b^3-a^3}{3}) \\
&= x^2 - (b+a)x + \frac{b^2+ab+a^2}{3}
\end{aligned}$$

and

$$\begin{aligned}
\overline{S}_1(T_1(U_{[a,b]}),U_{[a,b]}) &= \int_{\mathbb{R}} (\frac{a+b}{2} - y)^2 dU_{[a,b]}(y) \\
&= \int_a^b (\frac{(a+b)^2}{4} - 2\frac{a+b}{2}y + y^2)\frac{dy}{b-a} \\
&= \frac{1}{b-a}(\frac{(a+b)^2(b-a)}{4} - (a+b)\frac{b^2-a^2}{2} + \frac{b^3-a^3}{3}) \\
&= \frac{(a+b)^2}{4} - \frac{(a+b)^2}{2} + \frac{b^2+ab+a^2}{3}.
\end{aligned}$$

We see that $x^2 - (b+a)x + \frac{b^2+ab+a^2}{3}$ is a quadratic equation in x, which attains its minimum at $x = \frac{a+b}{2}$. Thus it holds that $\overline{S}_1(x, F) \geq \overline{S}_1(T_1(F), F)$ $\forall x \in \mathrm{A} = \mathbb{R}$ and $U_{[a,b]} \in \mathcal{F}_{\text{unif}}$. We have thus shown that S_1 is $\mathcal{F}_{\text{unif}}$-consistent with the mean. As the minimum is unique, we even get that S_1 is strictly $\mathcal{F}_{\text{unif}}$-consistent for the mean.

This does not come as surprise, given that it is an elementary result in probability theory that the mean $t := T_1(F)$ is the unique minimiser of $\mathbb{E}_F[(Y-x)^2]$. Not only is this property not unique to the family of uniform distributions but also easily generalisable to functionals other than the mean.

Lemma 2.4 (Example 2.4 in Brehmer and Strokorb [2019]). *Let $A \subseteq \mathbb{R}$ and let $g : O \to \mathbb{R}$ be such that g and g^2 are $\mathcal{F}$-integrable. Then $S_g(x, y) := (x - g(y))^2$ is strictly $\mathcal{F}$-consistent for $T_g : \mathcal{F} \to A,\ F \mapsto \overline{g}(F)$.*

Proof: Firstly, we compute

$$\begin{aligned}
\overline{S}_g(x, F) &= \int_{\mathrm{O}} (x - g(y))^2 dF(y) \\
&= \int_{\mathrm{O}} x^2 dF(y) - \int_{\mathrm{O}} 2xg(y)dF(y) + \int_{\mathrm{O}} g(y)^2 dF(y) \\
&= x^2 - 2x \int_{\mathrm{O}} g(y)dF(y) + \int_{\mathrm{O}} g(y)^2 dF(y).
\end{aligned}$$

Secondly, it holds that

$$\begin{aligned}
\overline{S}_g(T(F), F) &= \int_{\mathrm{O}} (\overline{g}(F) - g(y))^2 dF(y) \\
&= \int_{\mathrm{O}} \overline{g}(F)^2 dF(y) - \int_{\mathrm{O}} 2\overline{g}(F)g(y)dF(y) + \int_{\mathrm{O}} g(y)^2 dF(y) \\
&= \overline{g}(F)^2 - 2\overline{g}(F) \int_{\mathrm{O}} g(y)dF(y) + \int_{\mathrm{O}} g(y)^2 dF(y)
\end{aligned}$$

as $\overline{g}(F)$ is just a constant and $\int_{\mathrm{O}} 1dF(y) = 1$ as F is a probability measure. Once again, $x^2 - 2x \int_{\mathrm{O}} g(y)dF(y) + \int_{\mathrm{O}} g(y)^2 dF(y)$ attains its unique minimum at

$$x = \int_{\mathrm{O}} g(y)dF(y) = \overline{g}(F).$$

□

As remarked in Fissler and Ziegel [2016, Lemma 2.9], we can generally see that S is strictly $\mathcal{F}$-consistent for T if and only if for all $F \in \mathcal{F}$ the map $x \mapsto S(x, F)$ has a unique global minimum at $x = T(F)$.

2.3 Elicitability

In applications, the scoring function S often does not arise naturally when we think about a property of a distribution via a functional T but rather has to be chosen by the statistician. The question then becomes whether we can find a strictly $\mathcal{F}$-consistent scoring function S for T, i.e. if we can find S such that T minimises S in expectation. In general, the answer is no, as we will see later.

Functionals that do have such a scoring function are thus somehow special. This gives rise to the following definition.

Definition 2.5. *A functional $T : \mathcal{F} \to A \subseteq \mathbb{R}^n$ is called elicitable if there exists a strictly $\mathcal{F}$-consistent scoring function S for T. It is called jointly elicitable with $T' : \mathcal{F} \to A' \subseteq \mathbb{R}^k$ if $(T, T') : \mathcal{F} \to A \times A', F \mapsto (T(F), T'(F))$ is elicitable.*

Let $T_k(F) := \int_{\mathrm{O}} y^k dF(y)$, i.e. the functional mapping a distribution to its k-th moment. From what we derived in Lemma 2.4, in the case $\mathrm{O} \subseteq \mathbb{R}$, T_k is elicitable with $S_k(x, y) := (x - y^k)^2$. In our proof, we need that T_k is defined on a set of distributions which attain the $2k$-th moment. This is unsatisfying but indeed easy to fix. Brehmer [2017, Example 1.8] gives an approach to do this for the mean and uses it for the second moment in Example 1.23. We now use this approach for a more general class of functionals.

Example 2.6. Let $\mathrm{A} \subseteq \mathbb{R}^n$ and $\mathrm{O} \subseteq \mathbb{R}^d$ be sets, $f : \mathrm{A} \to \mathbb{R}$ be a strictly convex differentiable function and let $g : \mathrm{O} \to \mathbb{R}$ be $\mathcal{F}$-integrable. Then, the functional $T_g : \mathcal{F} \to \mathbb{R}, F \mapsto \overline{g}(F)$ is elicitable with the strictly $\mathcal{F}$-consistent scoring function $S(x, y) = -f(x) - \nabla f(x)^\top (g(y) - x)$.

Proof: Let $x \in \mathrm{A}$, $F \in \mathcal{F}$ and $t := T_g(F)$. Then, linearity of the integral and the definition of T_g yield the following:

$$\begin{aligned}
\overline{S}(x, F) - \overline{S}(t, F) &= \int_{\mathrm{O}} S(x, F) dF(y) - \int_{\mathrm{O}} S(t, F) dF(y) \\
&= \int_{\mathrm{O}} -f(x) - \nabla f(x)^\top (g(y) - x) dF(y) \\
&\quad - \int_{\mathrm{O}} -f(t) - \nabla f(t)^\top (g(y) - t) dF(y) \\
&= f(t) - f(x) - \nabla f(x)^\top \left(\int_{\mathrm{O}} g(y) dF(y) - x \right) \\
&\quad + \nabla f(t)^\top \left(\int_{\mathrm{O}} g(y) dF(y) - t \right) \\
&= f(t) - f(x) - \nabla f(x)^\top (T_g(F) - x) + \nabla f(t)^\top (T_g(F) - t) \\
&= f(t) - f(x) - \nabla f(x)^\top (t - x) \geq 0.
\end{aligned}$$

This last inequality follows from the strict convexity of f: $f(t) - f(x) \geq \nabla f(x)^\top (t - x)$ for all $x, t \in \mathrm{A}$ with equality if and only if $x = t = T_g(F)$. We get that S is strictly $\mathcal{F}$-consistent for T, which thus is elicitable. □

We see that in the case of moments, we do not need to assume the $2k$-th moment to be finite as we only integrate over y^k, whereas in Lemma 2.4, we have to integrate over y^{2k}.

Example 2.7. Gneiting and Raftery [2007, Section 6.1] show for $\mathrm{A}, \mathrm{O} \subseteq \mathbb{R}$ that the functional mapping a distribution to its α-quantile is elicitable with strictly consistent scoring function $S(x, y) = (\mathbb{1}(y \leq x) - \alpha)(x - y)$. This result assumes that $\mathcal{F}$ only contains distributions with strictly increasing cumulative distribution function and finite first moment.

One may then ask which functionals are not elicitable and the surprising answer is the variance. Although all moments are elicitable, the variance is not in general, as it fails the following criterion for elicitable functionals as we demonstrate in Example 2.9. Note that for $\lambda \in (0, 1)$ and $F_0, F_1 \in \mathcal{F}$, $\lambda F_1 + (1 - \lambda) F_0$ is again a probability distribution on $(\mathrm{O}, \mathcal{O})$, which can be interpreted as in Remark 3.1.

Theorem 2.8 (Convexity of level sets, Theorem 2.3 in Brehmer and Strokorb [2019])**.** *Let $T : \mathcal{F} \to A$ be an elicitable functional, $\lambda \in (0, 1)$ and $F_0, F_1 \in \mathcal{F}$. If $F_\lambda := \lambda F_1 + (1 - \lambda) F_0 \in \mathcal{F}$ and $T(F_0) = T(F_1) =: t$, then $T(F_\lambda) = t$.*

Proof: As T is elicitable, there exists a strictly consistent scoring function S for T. This means that for all $x \in \mathrm{A}$ the inequality $\overline{S}(T(F_\lambda), F_\lambda) \leq \overline{S}(x, F_\lambda)$ holds. Expanding and rearranging the right hand side yields

$$\begin{aligned}
\overline{S}(T(F_\lambda), F_\lambda) &\leq \overline{S}(x, F_\lambda) \\
&= \int_{\mathrm{O}} S(x,y) dF_\lambda(y) \\
&= \int_{\mathrm{O}} S(x,y) d(\lambda F_0 + (1-\lambda)F_0)(y) \\
&= \lambda \int_{\mathrm{O}} S(x,y) dF_0(y) + (1-\lambda) \int_{\mathrm{O}} S(x,y) dF_0(y) \\
&= \lambda \overline{S}(x, F_1) + (1-\lambda)\overline{S}(x, F_0).
\end{aligned}$$

By consistency of S with T, the right-hand side is minimised for $x = T(F_0) = T(F_1) = t$. As $T(F_\lambda) \in \mathrm{A}$, we get that $\overline{S}(T(F_\lambda), F_\lambda) = \overline{S}(t, F_\lambda)$ and as S is strictly consistent, $T(F_\lambda) = t$ must hold. □

A set of distributions $\mathcal{F}' \subseteq \mathcal{F}$ is said to be a level set for a functional T if $T(\mathcal{F}')$ is a singleton, i.e. T is constant on $\mathcal{F}'$. This theorem now tells us that a level set for an elicitable functional is convex.

Example 2.9 (Example 2.4 in Brehmer and Strokorb [2019])**.** The variance functional defined as $T_{\text{var}}(F) := T_2(F) - T_1^2(F)$ fails this condition in the following counterexample: Let $F_0 = U_{[-1,0]}$, $F_1 = U_{[0,1]}$ and $\lambda = \frac{1}{2}$, then $F_\lambda = U_{[-1,1]}$. We see that if T_{var} is the variance functional, $T_{\text{var}}(F_0) = T_{\text{var}}(F_1) = \frac{1}{12}$ but $T_{\text{var}}(F_\lambda) = \frac{1}{6}$.

Although the variance itself is not elicitable, it is jointly elicitable with the first moment.

Proof: (T_1, T_2) is elicitable with the strictly consistent scoring function $S_{1,2}((u,v),y) := (u-y)^2 + (v-y^2)^2$, which is just the sum of the two strictly consistent scoring functions S_1 and S_2 for T_1 and T_2 respectively. This follows as the unique minimiser (u,v) of the joint expected score is exactly the vector comprised of the two unique minimisers of the summands, which are just the expected scores of the two functionals.

Then, (T_1, T_{var}) is in bijection with (T_1, T_2) via $f((T_1, T_2)) = (T_1, T_2 - T_1^2) = (T_1, T_{\text{var}})$. We thus have that $S_{1,\text{var}}((u,v),y) := S_{1,2}(f^{-1}(u,v),y)$ is a strictly consistent scoring function for (T_1, T_{var}). □

From the definition it also follows that on a subset of $\mathcal{F}$, where T_1 is constant, T_{var} is just a shifted version of T_2. On this subset, T_{var} is elicitable. There is a special notion for this case as we define it now.

Definition 2.10. *Let $T : \mathcal{F} \to A \subseteq \mathbb{R}^n$ and $T' : \mathcal{F} \to A' \subseteq \mathbb{R}^k$ be functionals, and let T' be elicitable. For $x \in A'$, define $\mathcal{F}_x := \{F \in \mathcal{F} \mid T'(F) = x\}$. Then T is called conditionally elicitable given T' if for all $x \in A'$, $T|_{\mathcal{F}_x}$ is elicitable.*

Remark 2.11. As remarked by Brehmer and Strokorb [2019], the variance functional tells us that neither joint nor conditional elicitability imply elicitability. On the other hand, we can say that if a functional T is jointly elicitable with T' and T' itself is elicitable, then T is conditionally elicitable given T'. This just follows from the fact that the strictly consistent scoring function S from the joint elicitability of T and T' still is a strictly consistent scoring function when $T'(F)$ is constant.

2.4 Elicitation complexity

We see that it is somehow more complex to elicit the variance as we need it to be accompanied by another functional. The formalisation of this notion is captured in the following two definitions.

Definition 2.12. *$\mathcal{E}_k(\mathcal{F})$ is the set of $\mathbb{R}^k$-valued elicitable functionals defined on $\mathcal{F}$.*

Definition 2.13. *For a functional $T : \mathcal{F} \to A \subseteq \mathbb{R}$ and sets $\mathcal{C}_k \subseteq \mathcal{E}_k(\mathcal{F})$, the elicitation complexity of T with respect to $(\mathcal{C}_k)_{k\in\mathbb{N}}$ is defined by*

$$\mathit{elic}(T) := \min\{k \in \mathbb{N} \mid \exists\, T' \in \mathcal{C}_k : T = f \circ T' \text{ for some } f : T'(\mathcal{F}) \to A\}$$

and it is ∞ if there does not exist such $k \in \mathbb{N}$.

Remark 2.14. As demonstrated by Frongillo and Kash [2020, Remark 4], it makes sense to choose $\mathcal{C}_k \subsetneq \mathcal{E}_k(\mathcal{F})$ because else all functionals have elicitation complexity 1. They argue as follows. Take $\mathrm{O} = \mathbb{R}$ and note that a distribution F on $(\mathrm{O}, \mathcal{O})$ is uniquely determined by its cumulative distribution function (cdf), which in turn is uniquely defined by its values on the rationals $\mathbb{Q}$. Remember that the cdf is right continuous, so this uniqueness is not up to null sets. Define $g : \mathcal{F} \to \mathbb{R}^{\mathbb{Q}}$ to be the map that takes a distribution and evaluates its cdf on the rationals. Next, note that there is a bijection $\phi : \mathbb{R} \to \mathbb{R}^{\mathbb{Q}}$ as $\mathbb{R}$ and $\mathbb{R}^{\mathbb{Q}}$ have the same cardinality. We see that $h := \phi^{-1} \circ g : \mathcal{F} \to \mathbb{R}$ is injective as both g and ϕ^{-1} are, so there exists $h^{-1} : h(\mathcal{F}) \to \mathcal{F}$. Then, for a strictly proper scoring rule S^* as defined in Section 5, the scoring function $S(x, y) = S^*(h^{-1}(x), y)$ elicits $T' := h$, which means $T' \in \mathcal{E}_1(\mathcal{F})$. Finally, we can take $f := T \circ h^{-1}$ and see that $f \circ T' = T \circ h^{-1} \circ h = T$ and that $\mathrm{elic}(T) = 1$.

Possible choices of $(\mathcal{C}_k)_{k\in\mathbb{N}}$ include that $\mathcal{C}_k$ contains all bounded linear functionals or that the functionals are bounded with a differentiable, Lipschitz-continuous scoring function. It is also popular to require the functionals to have elicitable components, which means if $T' : \mathcal{F} \to \mathbb{R}^k$, $F \mapsto (T_{(1)}(F), \ldots, T_{(k)}(F))^\top$, then all component functionals $T_{(i)}, i \in \{1, 2, \ldots, k\}$ are elicitable. See Frongillo and Kash [2020] for more examples.

Counter-intuitively, we see that we cannot simply say that $\mathrm{elic}(T_{\mathrm{var}}) = 2$ without considering which regularity conditions we impose via $(\mathcal{C}_k)_{k\in\mathbb{N}}$. Luckily, Frongillo and Kash [2020, Section 3.2] show that with respect to a wide range of classes $(\mathcal{C}_k)_{k\in\mathbb{N}}$, we can actually say that $\mathrm{elic}(T_{\mathrm{var}}) = 2$.

2.5 Interpretation

This section gives an interpretation of the defined concepts and is inspired by Brehmer [2017, Section 2.1.2].

Take a decision maker, who wants to have the best prediction of a certain property of a random variable (or vector) Y that has an unknown distribution $F \in \mathcal{F}$. In a first test phase, the decision maker recruits forecasters, who hold opinions about F (potentially the whole distribution or else just some properties of F) and are tasked with issuing forecasts lying in A. The decision maker then assesses the forecasts based on some realisations and in the end chooses the best forecaster to issue future forecasts.

One option the decision maker has, is to tell the forecasters with which scoring function S their forecasts will be evaluated. A smaller value of the scoring function will then mean a bigger reward/ more points/ higher ranking for the forecasters. If $\overline{S}(\cdot, F')$ has a unique minimum for every $F' \in \mathcal{F}$, the forecasters will want to report the minimum of $\overline{S}(\cdot, F_i)$, where F_i is forecaster i's opinion of the underlying distribution.

On the other hand, the decision maker can communicate the functional T, i.e. which property of the distribution the forecasters should forecast. If T is elicitable, take S to be (one of) its strictly consistent scoring function(s). Naturally, the forecasters will predict $T(F_i)$, where again F_i is their opinion of the underlying distribution because $\overline{S}(\cdot, F_i)$ is minimised with this choice and they think that the decision maker uses S to assess their forecasts.

In this second example, elicitability of T is important to guarantee that the forecasters submit their real belief. If T is not elicitable, then the decision maker cannot choose a scoring function S to reliably assess the forecasts. If S is a consistent scoring function for T which is not strictly consistent, a forecaster might submit another minimum of the expected score and the decision maker would think it is a good prediction. If S is not consistent for T, then even predicting the correct value would not minimise the expected score, so a forecaster could submit a different value to get a lower score and thus a higher reward.

Note that the two options are equivalent and the decision maker can decide depending on the application which of the two makes more sense. Brehmer [2017, Section 2.1.2] mentions that this is exactly why such T are called elicitable. They "enable the decision maker to *elicit* truthful reports of the functional value."

Example 2.15. Imagine a construction company that installs solar panels in a city and wants to be able to give their prospective customers information about the distribution of the power they can generate. They model the light intensity in their city within a day in a specific month (e.g. September) as a random variable Y. Assume that Y has the unknown distribution $Y \sim F \in \mathcal{F}$, where $\mathcal{F}$ is a family of distributions with finite fourth moment. The construction company now asks forecasters (e.g. meteorologists) for forecasts to assess how good they are and in the end will choose one meteorologist to issue future forecasts. Let's say that meteorologist i uses some method to get a probability distribution $F_i \in \mathcal{F}$ for the light intensity.

Now the decision maker, i.e. the construction company, wants to know the expected amount of sun intensity on a day in September. There are two options: communicate that the forecasters should name their expected amount of light intensity or communicate that the decision maker will determine the score with $S_1(x, y) = (x - y)^2$, where x is the forecast and y is the actual observation next week.

In both cases, the forecasters will submit their true belief of the mean of the distribution; In the first case, the forecasters know that the mean is elicitable and thus the decision maker will use a strictly consistent scoring function to assess their forecasts. To maximise the expected reward, they thus report the minimiser of the strictly consistent scoring function, which is $T_1(F_i)$. In the second case, the forecasters also want to minimise their expected score, so they report the expected value of their distribution.

In another scenario, the construction company wants to know about the uncertainty in the light intensity via the variance of Y.

This, however, creates some problems. If the decision maker communicates that the forecasters should report the variance of their distribution, then, as the forecasters know that the variance functional is not elicitable, some may not submit a truthful value to try to minimise the score. On the other hand, assume the decision maker communicates the scoring function $S_2(x, y) = (x - y^2)^2$, with which the predictions are assessed. Note that the decision maker cannot choose the "right" scoring function as the variance functional is not elicitable. Then, the forecasters are incentivised to submit $T_2(F_i)$ instead of $T_{\text{var}}(F_i)$ to maximise their expected reward as S_2 is strictly consistent for T_2. In both cases, the decision maker may not receive the honest opinions of the forecasters and thus cannot make decisions based on the submitted information.

The way to fix this issue is that the decision maker asks the forecasters for the expected value **and** the variance of the light intensity next week as a pair (i.e. jointly). Then this pair is elicitable with the scoring function given in Example 2.9 and truthful reports are encouraged.

Chapter 3

Max-functionals

This chapter focuses on max-functionals, which are of central interest for the main take-away of this thesis. As they make use of convex combinations of probability distributions, from now on, we only consider sets $\mathcal{F}$ which are convex. Furthermore, we only look at univariate distributions, i.e. $\mathrm{O} \subseteq \mathbb{R}$, as they have a natural way to compare tail behaviour as we will see. We also identify a distribution F with its cumulative distribution function and we write $F(x)$ for the latter. Another notational detail is that we denote by $\overline{F}$ the survival function of the distribution F, which is defined as $\overline{F}(x) := 1 - F(x)$. This is not to be confused with the other bar notation we introduced, as they have nothing to do with each other.

3.1 Definition

Remark 3.1. Convex combinations of distributions have quite a nice interpretation. Given $F_0, F_1 \in \mathcal{F}$ and $\lambda \in (0,1)$, we can interpret sampling from $F_\lambda := \lambda F_1 + (1-\lambda)F_0$ as follows: we flip a coin which has a probability of λ of landing heads up and if it does, we independently of the coin flip sample from F_1 and else from F_0.

Brehmer [2017, Example 1.18] uses the following construction to show this statement. Let $Y_0 \sim F_0$, $Y_1 \sim F_1$ and $B \sim \mathcal{B}\text{ernoulli}(\lambda)$ independent of Y_0 and Y_1 and define $Y_\lambda := (1-B)Y_0 + BY_1$. By the law of total probability, it holds that

$$\begin{aligned}
\mathbb{P}(BY_0 + (1-B)Y_1 \leq x) &= \mathbb{P}((1-B)Y_0 + BY_1 \leq x, B = 0) \\
&\quad + \mathbb{P}((1-B)Y_0 + BY_1 \leq x, B = 1) \\
&= \mathbb{P}(Y_0 \leq x)\mathbb{P}(B = 0) + \mathbb{P}(Y_1 \leq x)\mathbb{P}(B = 1) \\
&= (1-\lambda)F_0(x) + \lambda F_1(x) = F_\lambda(x).
\end{aligned}$$

We thus see that $Y_\lambda \sim F_\lambda$ and that F_λ is indeed also a probability distribution on $(\mathrm{O}, \mathcal{O})$.

Definition 3.2. *A functional $T : \mathcal{F} \to \mathbb{R}$ is called a max-functional if for all $\lambda \in (0,1)$ and $F_0, F_1 \in \mathcal{F}$*

$$T(\lambda F_1 + (1-\lambda)F_0) = \max(T(F_0), T(F_1))$$

holds.

If a non-constant functional T is a max-functional, this means that the property of $\lambda F_1 + (1-\lambda)F_0$ we are looking at (via T) is dominated by (at least) one of the two distributions F_0 or F_1.

3.2 An example

While constant functionals are max-functionals, they do not tell us anything about a distribution. We instead look at some examples that may come up in applications.

Example 3.3. The upper endpoint x^F of a distribution $F \in \mathcal{F}$ is defined as

$$x^F = \sup\{x \in \mathbb{R} \mid F(x) < 1\}.$$

Then, $T : \mathcal{F}_{<\infty} \to \mathbb{R}, F \mapsto x^F$ is a max-functional on $\mathcal{F}_{<\infty} := \{F \mid x^F < \infty\}$. This just follows from the observation that

$$(\lambda F_1 + (1-\lambda)F_0)(x) = \lambda F_1(x) + (1-\lambda)F_0(x) < 1 \ \forall \lambda \in (0,1) \Leftrightarrow F_0(x) < 1 \text{ or } F_1(x) < 1,$$

which also immediately implies that $\mathcal{F}_{<\infty}$ is convex.

On an intuitive level, this just means that if our drawn sample can come from either of two distributions, the highest value we can get is the maximum of the highest value we can get from the given distributions.

While the upper endpoint is not suited to distinguish between distributions that can take arbitrarily large values, we introduce another characteristic that might be able to.

Definition 3.4 (Section 4 in Brehmer and Strokorb [2019])**.** *Let $F, G \in \mathcal{F}$ with upper end points of $x^F, x^G \in \mathbb{R} \cup \{\infty\}$ respectively. The strict partial order $<_t$ on $\mathcal{F}$ is defined as follows:*

$$F <_t G \textit{ if either } x^F < x^G \textit{ or if } x^F = x^G =: x^* \textit{ and } \lim_{x \to x^*} \frac{\overline{F}(x)}{\overline{G}(x)} = 0.$$

If $F <_t G$, we say that G has heavier tail than F.

The equivalence relation $\sim_t$ on $\mathcal{F}$ is defined as follows:

$$F \sim_t G \textit{ if } x^F = x^G =: x^* \textit{ and } \lim_{x \to x^*} \frac{\overline{F}(x)}{\overline{G}(x)} \in (0, \infty).$$

If $F \sim_t G$, we say that F and G are tail equivalent.

We omit the proofs that $<_t$ and $\sim_t$ are a strict partial order and an equivalence relation respectively as they follow from the definitions.

3.3 Further observations on max-functionals

This section explores some observations on max-functionals but is not relevant for the main take away of this thesis. We introduce some more max-functionals which usually are not of big interest. Another relevant example will then appear in Chapter 5.

Proposition 3.5 (Proposition 4.1 in Brehmer and Strokorb [2019])**.** *Let $T : \mathcal{F} \to \mathbb{R}$ be a functional such that for all $F, G \in \mathcal{F}$*

$$T(F) - T(G) \begin{cases} \leq 0 & \textit{if } F <_t G \\ \geq 0 & \textit{if } G <_t F \\ = 0 & \textit{else} \end{cases}$$

holds, then T is a max-functional.

Proof: Let $F_0, F_1 \in \mathcal{F}$ and for λ define $F_\lambda = \lambda F_1 + (1-\lambda)F_0$. We distinguish three cases. Firstly, if $F_0 <_t F_1$, we have that $x^{F_\lambda} = x^{F_1} \geq x^{F_0}$ as the upper endpoint is a max-functional and for $x < x^{F_1}$ it holds that

$$\frac{\overline{F_\lambda}(x)}{\overline{F_1}(x)} = \frac{\lambda\overline{F_1}(x) + (1-\lambda)\overline{F_0}(x)}{\overline{F_1}(x)} = \lambda + (1-\lambda)\frac{\overline{F_0}(x)}{\overline{F_1}(x)} \geq \lambda > 0.$$

We get $F_\lambda \sim_t F_1$ and thus neither $F_\lambda <_t F_1$ nor $F_1 <_t F_\lambda$ can hold as $<_t$ is a strict partial order. By the assumptions on T, we can conclude by $T(F_\lambda) = T(F_1) \geq T(F_0)$. The case $F_1 <_t F_0$ can be treated analogously by symmetry.

The third case is if neither $F_0 <_t F_1$ nor $F_1 <_t F_0$ holds, so $x^{F_0} = x^{F_1} = x^{F_\lambda} =: x^*$. Consequently,

$$\liminf_{x\to x^*} \frac{\overline{F_\lambda}(x)}{\overline{F_1}(x)} \geq \lambda > 0 \text{ and } \limsup_{x\to x^*} \frac{\overline{F_\lambda}(x)}{\overline{F_1}(x)} < \infty,$$

where the inequalities come from above expansion and the fact that F_0 does not have heavier tail than F_1. We can thus say that neither $F_\lambda <_t F_1$ nor $F_1 <_t F_\lambda$ and by the assumptions on T, $T(F_\lambda) = T(F_1) = T(F_0)$. □

Note that this Proposition is not an if and only if statement. In essence, the backwards direction does not need to hold because a max-functional may very well be derived from the negative of a min-functional which orders the lower tails but contains no information on the upper tails.

Example 3.6. Consider $T(F) = x_F$, where x_F is the lower endpoint of the distribution F, which is defined analogously to the upper endpoint. We see that T is a min-functional, so for all $\lambda \in (0,1)$ it holds that $T(\lambda F_1 + (1-\lambda)F_0) = \min\{T(F_0), T(F_1)\}$ by an analogous argument as in Example 3.3. This directly implies that $T'(F) = -T(F)$ is a max-functional as

$$\begin{aligned} T'(\lambda F_1 + (1-\lambda)F_0) &= -T(\lambda F_1 + (1-\lambda)F_0) \\ &= -\min\{T(F_0), T(F_1)\} \\ &= \max\{-T(F_0), -T(F_1)\} \\ &= \max\{T'(F_0), T'(F_1)\}. \end{aligned}$$

As a concrete counterexample, take $F = U_{[-1,0]}$, $G = U_{[0,1]}$ and $T'(F) = -x_F$. Then we have that $T'(F) - T'(G) = 1 - 0 \geq 0$ although $F <_t G$.

From this example, one may get the intuition that being a max-functional is a property of the upper or lower tail. This intuition is wrong as the next example tells us.

Example 3.7. Let $\mathcal{F}_{\text{mid}} := \{F \in \mathcal{F}^* \mid F(0) = \frac{1}{2}\}$, i.e. distributions where the origin is contained in the set of their median(s). $\mathcal{F}_{\text{mid}}$ is convex as for $F_0, F_1 \in \mathcal{F}_{\text{mid}}$ and $\lambda \in (0,1)$, we have $\lambda F_1(0) + (1-\lambda)F_0(0) = \lambda\frac{1}{2} + (1-\lambda)\frac{1}{2} = \frac{1}{2}$. We then define $T : \mathcal{F}_{\text{mid}} \to \mathbb{R}$, to be the functional such that

$$T(F) = \sup\{x \in (-\infty, 0) \mid F(x) < \frac{1}{2}\}.$$

Let us now show that T is a max functional. Take $F_0, F_1 \in \mathcal{F}_{\text{mid}}$, $\lambda \in (0,1)$ and assume that $T(F_0) < T(F_1)$, then, for $x \in [T(F_0), T(F_1))$

$$\lambda F_1(x) + (1-\lambda)F_0(x) = \lambda F_1(x) + (1-\lambda)\frac{1}{2} < \frac{1}{2}$$

and we see that if $x = T(F_1), F_\lambda(x) = \frac{1}{2}$. This means that $T(F_\lambda) = T(F_1)$. The case if $T(F_1) < T(F_0)$ is analogous and if $T(F_0) = T(F_1)$ it is clear that $T(F_\lambda) = T(F_0) = T(F_1)$, concluding the proof that T is a max-functional.

We see that this max-functional has nothing to do with the tails as one can easily construct distributions of the form $F = \frac{1}{2}U_{[a,b]} + \frac{1}{2}U_{[c,d]}$ and $G = \frac{1}{2}U_{[a',b']} + \frac{1}{2}U_{[c',d']}$ where $a < b \leq 0 \leq c < d$ and $a' < b' \leq 0 \leq c' < d'$ such that $T(F) - T(G)$ has any sign and F and G are upper or lower tail ordered in any way.

Example 3.8. Another example of a max-functional is

$$T : \mathcal{F} \to \{0, 1\}, F \mapsto \mathbb{1}(\lim_{x \uparrow 0} F(x) \neq F(0))$$

i.e. the indicator function of whether F has a point mass at 0. For $F_0, F_1 \in \mathcal{F}$ and $\lambda \in (0, 1)$, let us look at

$$\lim_{x \uparrow 0} F_\lambda(x) = \lim_{x \uparrow 0} \lambda F_1(x) + (1 - \lambda)F_0(x).$$

If F_0 or F_1 have a jump at $x = 0$, this quantity does not equal $\lambda F_1(0) + (1 - \lambda)F_0(0)$ and thus $T(F_\lambda) = 1 = \max(T(F_0), T(F_1))$. If neither F_0 nor F_1 have a jump at $x = 0$, then $F_\lambda(x)$ is continuous at $x = 0$, so $T(F_\lambda) = 0 = \max(0, 0)$.

While we know now that being a max-functional does not mean a functional orders the upper or lower tail in any way, we can say that they usually focus on low probability regions as demonstrated in Remark 3.11. Let us fist introduce a concept needed for this remark.

Definition 3.9. *For $0 \leq \alpha < \beta \leq 1$ and a distribution F, define*

$$F_{[\alpha,\beta]}(x) := \frac{F(x) - \alpha}{\beta - \alpha} \mathbb{1}(\alpha \leq F(x) < \beta) + \mathbb{1}(\beta \leq F(x)).$$

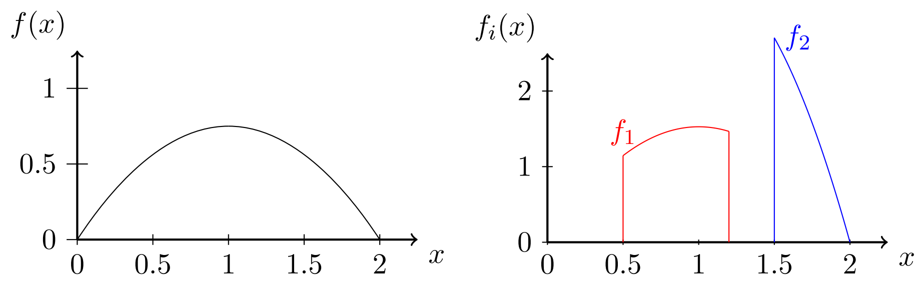
Let us call this the α, β-quantilised distribution of F.

Note that $F_{[\alpha,\beta]}$ is again a valid distribution as $\lim_{x\to-\infty} F_{[\alpha,\beta]}(x) = 0$, $\lim_{x\to\infty} F_{[\alpha,\beta]}(x) = 1$, $F_{[\alpha,\beta]}$ is increasing, is right continuous and has left limits. This is a consequence of F having those properties and because the points $F(x) = \alpha$ and $F(x) = \beta$ are in line with these requirements.

Example 3.10. For a distribution F attaining a density f, one can visualise the density $f_{[\alpha,\beta]}$ of $F_{[\alpha,\beta]}$ quite neatly as can be seen in the following diagrams. We plot the density

$$f(x) = \mathbb{1}(0 \leq x \leq 2)\frac{3}{4}x(2 - x)$$

in the first plot and the two new densities $f_1 = f_{[\frac{5}{32}, \frac{81}{125}]}$ and $f_2 = f_{[\frac{27}{32}, 1]}$ in the second plot. Note that the distributions are not simply cut out of the original distribution but need to be rescaled to be valid distributions again. We do not plot the distributions where they are zero and draw vertical lines at the boundary of the support for clarity.



Remark 3.11. Assume now $\mathcal{F}$ to be sufficiently rich in the sense that for every $0 \leq \alpha < \beta \leq 1$ and $F \in \mathcal{F}$, $F_{[\alpha,\beta]} \in \mathcal{F}$ and all convex combinations of such quantilised distributions are in $\mathcal{F}$. This assumption is for example satisfied if we consider $\operatorname{conv}(\mathcal{F}_{\text{unif}})$ or the set of all distributions with finite first moment.

In $\mathcal{F}$ it holds that for all $\lambda \in (0,1)$ and F, $F = \lambda F_{[0,\lambda]} + (1-\lambda)F_{[\lambda,1]}$. This means for a max-functional $T : \mathcal{F} \to \mathbb{R}$ that $T(F) = \max(T(F_{[0,\lambda]}), T(F_{[\lambda,1]}))$ for all $\lambda \in (0,1)$. In fact, for $n \in \mathbb{N}$, we can choose $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_{n+1} = 1$ and decompose $F = \sum_{i=0}^{n}(\lambda_{i+1} - \lambda_i)F_{[\lambda_i,\lambda_{i+1}]}$. By repeatedly applying the property that T is a max functional and as the maximum of maxima is the overall maximum, we get $T(F) = \max_{i\in\{0,1,\dots,n\}} T(F_{[\lambda_i,\lambda_{i+1}]})$.

In the case of the upper endpoint, the i that attains this maximum is always $i = n$ and in the case of the (negative of the) lower endpoint in Example 3.6 it is $i = 0$. For the max-functional in Example 3.8, it is the i such that $\lambda_i \leq F(0) < \lambda_{i+1}$.

In general, we cannot say that there is a unique i which attains this maximum as we can easily generalise Example 3.8 and say that the functional T is the indicator functional whether F has a point mass at $x = 0$ or $x = 1$. Similarly, we can extend it to m point masses and can have up to m maxima.

3.4 Properties of max-functionals

Given our framework of elicitability, one might ask (when) are max-functionals elicitable? The answer is quite simple; unless they are constant, never. The next question that arises is what their elicitation complexity is, if not one. Under mild assumptions, it is actually infinite.

Theorem 3.12 (Theorem 3.3 in Brehmer and Strokorb [2019])**.** *Let $T : \mathcal{F} \to A$ be a functional. Assume there are F_0, F_1 such that $T(F_0) \neq T(F_1)$ and $\forall \lambda \in (0,1)$ the property*

$$T(\lambda F_1 + (1-\lambda)F_0) \in \{T(F_0), T(F_1)\}$$

holds, then T is not elicitable.

Proof: Define $x_0 := T(F_0)$, $x_1 := T(F_1)$ and $F_\lambda := \lambda F_1 + (1-\lambda)F_0$. By contradiction, assume there exists a strictly consistent scoring function S for T. We then have by the linearity of the integral

$$\overline{S}(x_0, F_\lambda) - \overline{S}(x_1, F_\lambda) = \lambda(\overline{S}(x_0, F_1) - \overline{S}(x_1, F_1)) + (1-\lambda)(\overline{S}(x_0, F_0) - \overline{S}(x_1, F_0)).$$

Note that the first difference $\overline{S}(x_0, F_1) - \overline{S}(x_1, F_1)$ is positive and the second difference $\overline{S}(x_0, F_0) - \overline{S}(x_1, F_0)$ is negative as x_i is the minimiser for F_i by consistency of S. We can thus choose λ such that the right-hand side equals to 0 and for this λ we get $\overline{S}(x_0, F_\lambda) = \overline{S}(x_1, F_\lambda)$. By our restriction on T, either $T(F_\lambda) = x_0$ or $T(F_\lambda) = x_1$ must hold. As $T(F_\lambda)$ is the unique minimizer of $\overline{S}(\cdot, F_\lambda)$ and $x_0 \neq x_1$, we get a contradiction. □

Corollary 3.13 (Corollary 3.4 in Brehmer and Strokorb [2019])**.** *Non-constant max-functionals are not elicitable.*

Definition 3.14. *A functional $T : \mathcal{F} \to A$ is called mixture continuous if for all $F_0, F_1 \in \mathcal{F}$ the mapping*

$$[0,1] \to A,\ \lambda \mapsto T(\lambda F_1 + (1-\lambda)F_0)$$

is a continuous function.

Here we ask for continuity in the usual euclidean topologies of $[0,1]$ and $\mathrm{A} \subseteq \mathbb{R}^k$. Naturally, non-constant max-functionals are not mixture continuous.

Example 3.15. The mean functional is mixture continuous as are all the moment functionals. In fact, all integral functionals $T_g(F) = \overline{g}(F)$ in the sense of Lemma 2.4 are mixture continuous as

$$\begin{aligned} T(\lambda F_1 + (1-\lambda)F_0) &= \int_{\mathrm{O}} g(y) d(\lambda F_1 + (1-\lambda)F_0)(y) \\ &= \lambda \int_{\mathrm{O}} g(y) dF_1(y) + (1-\lambda) \int_{\mathrm{O}} g(y) F_0(y) \\ &= \lambda T(F_1) + (1-\lambda) T(F_0) \\ &= \lambda (T(F_1) - T(F_0)) + T(F_0), \end{aligned}$$

so the corresponding map $\lambda \mapsto T(\lambda F_1 + (1-\lambda)F_0)$ is an affine linear map and thus continuous. Furthermore, Fissler and Ziegel [2019, Section 2.1] show that quantiles are mixture continuous and more generally that an elicitable functional T' is mixture continuous if $x \mapsto \overline{S}(x, F)$ is a continuous map for all $F \in \mathcal{F}$ and if $T(\mathcal{F})$ is bounded.

Mixture continuity comes into play in the following definitions.

Definition 3.16. *We define the two classes $(\mathcal{U}_k)_{k \in \mathbb{N}}$ and $(\mathcal{V}_k)_{k \in \mathbb{N}}$ as follows:*

$$\mathcal{U}_k := \{T' \in \mathcal{E}_k(\mathcal{F}) \mid T' \textit{ is mixture-continuous, has elicitable components and } int(T'(\mathcal{F})) \neq \emptyset\},$$

where $int(T'(\mathcal{F}))$ denotes the (open) interior of the set $T'(\mathcal{F}) \subseteq \mathbb{R}^k$, and

$$\mathcal{V}_k := \{T' \in \mathcal{E}_k(\mathcal{F}) \mid T' \textit{ is mixture-continuous, has elicitable components and } T'(\mathcal{F}) \textit{ is open}\}.$$

These classes play a crucial role in understanding the elicitation complexity of max-functionals as the following theorem demonstrates.

Theorem 3.17 (Theorem 3.7 in Brehmer and Strokorb [2019])**.** *Let $T : \mathcal{F} \to \mathbb{R}$ be a max-functional. Then the following statements hold true.*

(i) $elic(T) = \infty$ with respect to $(\mathcal{U}_k)_{k \in \mathbb{N}}$ unless $T(\mathcal{F})$ contains its supremum.

(ii) $elic(T) = \infty$ with respect to $(\mathcal{V}_k)_{k \in \mathbb{N}}$ unless T is constant.

The proof for this theorem can be found in Brehmer and Strokorb [2019, Theorem 3.7] and is very elaborated, so we only give a sketch.

Proof: The proof argues that if a functional T' as in Definition 2.13 exists, then its image is path connected and we can choose a hyperrectangle Q in the interior of said image. The elicitable components of T' together with the convexity of level sets and mixture continuity imply that the link function f between T and T' is constant on $\mathrm{int}(Q)$.

Mixture continuity then implies that for any $F_0, F_1 \in \mathcal{F}$ such that $T'(F_0) \in \mathrm{int}(Q)$, there is μ small enough such that $T'(\mu F_1 + (1-\mu)F_0) \in \mathrm{int}(Q)$. Applying f to $T'(\mu F_1 + (1-\mu)F_0)$ and using that T is a max-functional yield that actually $f(x)$ is smaller than $f(\mathrm{int}(Q)$ for every x in the image of T', proving the first statement. The second statement follows as we can choose a hyperrectangle around every x in the image of T', on which f is again constant and thus f is constant on the whole image of T'. □

Corollary 3.18 (Corollary 3.8 in Brehmer and Strokorb [2019]). *Let $T : \mathcal{F} \to \mathbb{R}$ be a max-functional such that one of the following conditions is satisfied.*

(i) T is unbounded

(ii) T is surjective onto an open interval (a, b).

(iii) T is surjective onto a half-open interval $[a, b)$.

Then, elic$(T) = \infty$ with respect to $(\mathcal{U}_k)_{k\in\mathbb{N}}$.

We have thus shown that max-functionals have infinite elicitation complexity in a wide range of settings.

Example 3.19. Let $\mathcal{F}$ be the set of all probability distributions on $(\mathrm{O}, \mathcal{O})$, consider $\mathcal{F}_{<\infty}$ and $T : \mathcal{F}_{<\infty} \to \mathbb{R}, F \mapsto x^F$. As T is not constant, it has infinite elicitation complexity with respect to $(\mathcal{V}_k)_{k\in\mathbb{N}}$. Furthermore, it also has infinite elicitation complexity with respect to $(\mathcal{U}_k)_{k\in\mathbb{N}}$ as T is unbounded. This just follows from the fact that $x^{U_{[0,m]}} \to \infty$ as $m \to \infty$.

Theorem 3.20 (Theorem 3.9 in Brehmer and Strokorb [2019]). *Let $T : \mathcal{F} \to \mathbb{R}$ be a max-functional such that elic$(T) = \infty$ with respect to a family $(\mathcal{C}_k)_{k\in\mathbb{N}}$. Let $T' : \mathcal{F} \to \mathrm{A}'$ be a functional with $T' \in \mathcal{C}_m$ for some $m \in \mathbb{N}$. Then the following hold true.*

(i) T is not conditionally elicitable given T'.

(ii) T is not jointly elicitable with T'.

Proof: To prove the first part, assume by contradiction that T is elicitable on the subclass $\mathcal{F}_y = \{F \in \mathcal{F} \mid T'(F) = y\}$ for all $y \in \mathrm{A}'$.

First, let $\lambda \in (0, 1)$ and $F_0, F_1 \in \mathcal{F}_y$ for $y \in \mathrm{A}'$. Recall from the proof of 2.8 that if T' is elicitable, there is a strictly consistent scoring function S for T' and the equality $\overline{S}(x, F_\lambda) = \lambda\overline{S}(x, F_1) + (1 - \lambda)\overline{S}(x, F_0)$ holds. Then again $T'(F_\lambda) = T(\lambda F_1 + (1 - \lambda)F_0)$ is the unique minimiser of the left-hand side, which must also be the unique minimiser of the right-hand side, which is $T'(F_0) = T'(F_1) = z$. This means that $\lambda F_1 + (1 - \lambda)F_0 \in \mathcal{F}_y$ and $\mathcal{F}_y$ is thus convex for all $y \in \mathrm{A}'$.

By the assumption that elic$(T) = \infty$, there is no link function f such that $T = f \circ T'$ holds. This means that there is a $z \in \mathrm{A}'$ such that T is not constant on $\mathcal{F}_z$ as else we could define $f(T'(F)) := T(F)$.

By Theorem 2.8, we get that T cannot be elicitable on $\mathcal{F}_z$ as there exist $F_0, F_1 \in \mathcal{F}_z$ such that $T(F_0) \neq T(F_1)$.

The second part of this theorem follows from Remark 2.11, which says that joint elicitability implies conditional elicitability with T', and the first part of this theorem. $\square$

Chapter 4

Scoring rules

In light of Section 2.5, one might ask why the decision maker does not simply ask the forecasters to report the whole distribution F_i, which they think is responsible for the process. This approach seems reasonable given that the decision maker can then just read off the desired properties via a functional. Nevertheless, point forecasts are common in all areas of our day-to-day life, for one because they are easy to convey and to interpret. Furthermore, the decision maker still has to evaluate which is the best predicted distribution based on observed data.

In this chapter, we show that even when whole distribution functions are returned, so called probabilistic forecasting, the forecasts can differ substantially from the real distribution in terms of tail behaviour without being noticed.

4.1 Definition

The pendant of a scoring function in point forecasting is a scoring rule. Instead of a vector of predicted points from the action domain, it takes a distribution as input and compares it to actual realisations.

Definition 4.1. *$S : \mathcal{F} \times O \to \mathbb{R}$ is called a scoring rule if for all $F \in \mathcal{F}$ the mapping $y \mapsto S(F, y)$ is $\mathcal{F}$ integrable. S is called a proper scoring rule if $\overline{S}(F, F) \leq \overline{S}(G, F)$ $\forall F, G \in \mathcal{F}$. S is called strictly proper if for all $F, G \in \mathcal{F}$ the equality $\overline{S}(G, F) = \overline{S}(F, F)$ implies $G = F$.*

This definition just means that if a forecaster predicts the right distribution, then the expected score is the lowest. We continue with a notion of continuity for scoring rules, which will later be important for the main take-away of this chapter.

Definition 4.2. *A scoring rule $S : \mathcal{F} \times O \to \mathbb{R}$ is called diagonal continuous at $G \in \mathcal{F}$, if for all $F \in \mathcal{F}$*

$$\overline{S}(\lambda F + (1 - \lambda)G, G) \to \overline{S}(G, G) \text{ for } \lambda \downarrow 0.$$

4.2 Main result

Lemma 4.3 (Lemma 5.3 in Brehmer and Strokorb [2019])**.** *Let $S : \mathcal{F} \times O \to \mathbb{R}$ be a proper scoring rule, then it is diagonal continuous at each $G \in \mathcal{F}$.*

Proof: Let $F, G \in \mathcal{F}$, $\lambda \in [0,1)$ and define $F_\lambda := \lambda F + (1-\lambda)G$. We get

$$\begin{aligned}
(1-\lambda)\overline{S}(F_\lambda, G) &= (1-\lambda)\int_{\mathrm{O}} S(F_\lambda, y)dG(y) \\
&= (1-\lambda)\int_{\mathrm{O}} S(F_\lambda, y)dG(y) + \lambda\int_{\mathrm{O}} S(F_\lambda, y)dF(y) - \lambda\int_{\mathrm{O}} S(F_\lambda, y)dF(y) \\
&= \int_{\mathrm{O}} S(F_\lambda, y)dF_\lambda(y) - \lambda\int_{\mathrm{O}} S(F_\lambda, y)dF(y) \\
&= \overline{S}(F_\lambda, F_\lambda) - \lambda\overline{S}(F_\lambda, F) \\
&\leq \overline{S}(G, F_\lambda) - \lambda\overline{S}(F, F) \\
&= (1-\lambda)\overline{S}(G, G) + \lambda\overline{S}(G, F) - \lambda\overline{S}(F, F),
\end{aligned}$$

where the inequality follows from S being a strictly proper scoring rule and the last equality from de- and recomposing $\overline{S}(G, F_\lambda)$. Rearranging the terms gives

$$|\overline{S}(\lambda F + (1-\lambda)G, G) - \overline{S}(G, G)| \leq \frac{\lambda}{1-\lambda}(\overline{S}(G, F) - \overline{S}(F, F)),$$

which tends to zero as λ goes to zero. □

Diagonal continuity is of central importance for our next theorem, which is the main takeaway of this section. It says that no scoring rule can reliably differentiate between not tail equivalent distributions.

Theorem 4.4 (Theorem 5.4 in Brehmer and Strokorb [2019])**.** *Let $S : \mathcal{F} \times \mathbb{R} \to \mathbb{R}$ be a proper scoring rule and $G \in \mathcal{F}$, then the following hold.*

(i) If there is $F \in \mathcal{F}$ with heavier tail than G, then for all $\varepsilon > 0$ there is F_ε not tail equivalent to G such that

$$|\overline{S}(F_\varepsilon, G) - \overline{S}(G, G)| \leq \varepsilon.$$

(ii) Let $T : \mathcal{F} \to \mathbb{R}$ be a max-functional. If there is $F \in \mathcal{F}$ with $T(F) > T(G)$, then for all $\varepsilon > 0$ there is $F_\varepsilon \in \mathcal{F}$ such that $T(F_\varepsilon) = T(F) > T(G)$ while

$$|\overline{S}(F_\varepsilon, G) - \overline{S}(G, G)| \leq \varepsilon.$$

Proof: For $F, G \in \mathcal{F}$ and $\lambda \in [0,1]$, define $F_\lambda = \lambda F + (1-\lambda)G \in \mathcal{F}$. In case of (i), assume that F has heavier tail than G. If $x^F > x^G$, then $x^{F_\lambda} > x^G$ for all $\lambda \in (0,1]$ as the upper endpoint is a max-functional. If $x^F = x^G =: x^*$, we have

$$\frac{\overline{F_\lambda}(x)}{\overline{G}(x)} = \frac{\lambda\overline{F}(x) + (1-\lambda)\overline{G}(x)}{\overline{G}(x)} = (1-\lambda) + \lambda\frac{\overline{F}(x)}{\overline{G}(x)}$$

for $x < x^*$. The right-hand side of this equation goes to infinity as $x \to x^*$ because F has heavier tail than G. This means that either way F_λ is not tail equivalent to G for $\lambda \in (0,1]$. In case of (ii), assume $T(F) > T(G)$. As T is a max-functional, $T(F_\lambda) = T(F) > T(G)$ for all $\lambda \in (0,1]$.

Now, in both cases, by Lemma 4.3, S is diagonal continuous at G and thus for all $\varepsilon > 0$ there is $\delta \in (0,1]$ such that $|\overline{S}(F_\lambda, G) - \overline{S}(G, G)| \leq \varepsilon$ holds for all $\lambda \in [0, \delta]$. We can conclude both statements by choosing $F_\varepsilon = F_\lambda$ for some $\lambda \in (0, \delta]$. □

Remark 4.5. As we can usually not calculate $\overline{S}(G, F)$ because we do not know the real distribution F, we have to estimate $\overline{S}(G, F)$ by $\hat{S}(G, F) := \frac{1}{n}\sum_{i=1}^n S(G, y_i)$ for realisations y_i. Under appropriate conditions, we can apply the law of large numbers and know that $\frac{1}{n}\sum_{i=1}^n S(G, y_i)$ converges to $\overline{S}(G, F)$ as $n \to \infty$. This still creates some problems. Take a set $M = \{F_1, F_2, \ldots, F_n\}$

of forecasts we want to compare and then choose the $F_i \in M$ with the lowest $\hat{S}(F_i, F)$. If the real distribution F is $\in M$ we are not guaranteed to choose F, as $\hat{S}(F, F)$ might not be minimal as we evaluate $\hat{S}$ on a finite amount of data. Furthermore, by Theorem 4.4, we are not guaranteed to choose a functional tail-equivalent to F.

Chapter 5

Simulation study

We conclude this thesis with a small simulation study. First, we motivate and introduce a family of distributions $\mathcal{F}_{\text{exp}}$, which has a very natural notion of tail behaviour and tail equivalence. Then, we simulate some data to support Theorem 4.4.

5.1 Convex combinations of exponential distributions

The time elapsed between random events (so called interarrival or interevent time) often is modelled by a random variable Y having an exponential distribution. Such events include customers entering a store or earthquakes.

A big draw back of using the exponential distribution for modelling is that having only one parameter means the model usually is too simple. Especially if several factors come into play like different kinds of customers which enter a store at different rates, modelling the entirety with an exponential distribution is not justifiable. For this reason, Polymenis [2025] models the time between earthquakes happening as a mixture of exponential distributions.

We take this as an incentive to take $\mathrm{O} = [0, \infty)$ and define

$$\mathcal{F}_{\text{exp}} := \operatorname{conv}(\{E_\theta \sim \operatorname{Exp}(\theta) \mid \theta \in (0, \infty)\}),$$

the convex hull of the family of exponential distributions. Generally, by E_θ we refer to a the exponential distribution with parameter θ.

Remark 5.1. By the definition of the convex hull, every $G \in \mathcal{F}_{\text{exp}}$ is of the form:

$$G = \sum_{i=1}^{n} \lambda_i E_{\theta_i}, \text{ s.t. } \sum_{i=1}^{n} \lambda_i = 1, \lambda_i > 0, E_{\theta_i} \sim \operatorname{Exp}(\theta_i), \theta_i \in (0, \infty) \ \forall i \in \{1, 2, \dots, n\},$$

for some $n \in \mathbb{N} \setminus \{0\}$. For convenience, we order the θ_i such that $\theta_1 < \theta_2 < \cdots < \theta_n$.

On this set, there exists a natural way to define tail behaviour.

Definition 5.2. *For a distribution $G \in \mathcal{F}_{exp}$, define $T_\rho(G)$ to take the value $\rho \in (0, \infty)$ such that*

$$\lim_{x \to \infty} \frac{\overline{G}(x)}{e^{\rho x}} \in (0, \infty).$$

We prove existence and uniqueness of ρ starting with uniqueness.

Proof: If there are ρ, ρ' with this property, then

$$\lim_{x\to\infty} \frac{e^{\rho' x}}{\overline{G}(x)} \cdot \lim_{x\to\infty} \frac{\overline{G}(x)}{e^{\rho x}} = \lim_{x\to\infty} \frac{e^{\rho' x}}{\overline{G}(x)} \frac{\overline{G}(x)}{e^{\rho x}} = \lim_{x\to\infty} \frac{e^{\rho' x}}{e^{\rho x}} = \lim_{x\to\infty} e^{(\rho'-\rho)x} \in (0,\infty),$$

which can only happen if $\rho' = \rho$.

For every $G \in \mathcal{F}_{\text{exp}}$, there is such a ρ, as we can take the expansion of $G = \sum_{i=1}^n \lambda_i E_{\theta_i}$ for $E_{\theta_i} \sim Exp(\theta_i)$. Recall that $\theta_1 < \cdots < \theta_n$ and take $\rho = -\theta_1$, then

$$\lim_{x\to\infty} \frac{\overline{G}(x)}{e^{\rho x}} = \lim_{x\to\infty} \frac{\sum_{i=1}^n \lambda_i e^{-\theta_i x}}{e^{\rho x}} = \lim_{x\to\infty} \sum_{i=1}^n \lambda_i e^{-(\theta_i+\rho)x} = \sum_{i=1}^n \lim_{x\to\infty} \lambda_i e^{-(\theta_i+\rho)x} = \lambda_1 > 0$$

because all the summands of the i which are not minimising θ_i will converge to 0 because $\theta_i + \rho > 0$. □

Lemma 5.3. *The functional $T_\rho : \mathcal{F}_{exp} \to (0,\infty)$ is a max-functional.*

Proof: Let $F_0, F_1 \in \mathcal{F}_{\text{exp}}$, $\rho_0 := T_\rho(F_0)$ and $\rho_1 := T_\rho(F_1)$. For $\lambda \in (0,1)$, we look at $F_\lambda := \lambda F_1 + (1-\lambda)F_0$ and get that

$$\lim_{x\to\infty} \frac{\overline{F}_\lambda(x)}{e^{\rho x}} = \lim_{x\to\infty} \lambda \frac{\overline{F}_1(x)}{e^{\rho x}} + (1-\lambda)\frac{\overline{F}_0(x)}{e^{\rho x}}.$$

If we want this value to converge to a number in $(0,\infty)$, ρ has to equal $\max(\rho_0, \rho_1)$. If ρ were strictly bigger, both summands would converge to zero, so their sum would also converge to zero and if ρ were strictly smaller than the given maximum, at least one of the summands would diverge to infinity and so would their sum. Only with this choice of ρ both summands are guaranteed not to diverge and at least one of the summands is non-zero, so $T_\rho(F_\lambda) = \max(\rho_0, \rho_1) = \max(T_\rho(F_0), T_\rho(F_1))$ and T_ρ is thus a max-functional. □

Lemma 5.4. *For $G_0, G_1 \in \mathcal{F}_{exp}$ with expansions $G_0 = \sum_{i=1}^n \lambda_i E_{\theta_i}, G_1 = \sum_{i=1}^m \lambda_i' E_{\nu_i}$, it holds that*

$$\theta_1 = \nu_1 \Leftrightarrow G_0 \sim_t G_1.$$

Proof: We see that for $E_\theta \sim \text{Exp}(\theta), \overline{E}_\theta(x) = e^{-\theta x}$, so by comparing Definition 5.2 and Definition 3.4 (tail equivalence) for $G \in \mathcal{F}_{\text{exp}}$ it holds that $G \sim_t E_\theta \Leftrightarrow T_\rho(G) = \theta$. Hence, $G_0 \sim_t E_{\theta_1}$ and $G_1 \sim_t E_{\nu_1}$. We can conclude by the observation that tail equivalence is an equivalence relation in the first logical equivalence and our finding from before in the second logical equivalence:

$$G_0 \sim_t G_1 \Leftrightarrow E_{\theta_1} \sim_t E_{\nu_1} \Leftrightarrow T(E_{\theta_1}) = \nu_1 \Leftrightarrow \theta_1 = \nu_1.$$

□

5.2 CRPS

One popular example of a scoring rule is the weighted continuous ranked probability score or wCRPS for short. It is defined as follows:

$$\text{wCRPS}(F, y) := \int_{-\infty}^{\infty} w(x)(F(x) - \mathbb{1}(y \le x))^2 dx.$$

For the sake of simplicity, we use $w(x) = 1$ for all $x \in \mathbb{R}$, which is the so called continuous ranked probability score CRPS. It is a strictly proper scoring rule as for all families $\mathcal{F}$ only containing distributions that have finite first moment per Gneiting and Raftery [2007, Section 4.2].

For $G \in \mathcal{F}_{\text{exp}}$ and $y \in [0, \infty)$, we compute

$$\begin{aligned}
\text{CRPS}(G, y) &= \int_{-\infty}^{\infty} (G(x) - \chi(y \le x))^2 dx \\
&= \int_0^{\infty} (1 - \sum_{i=1}^{n} \lambda_i e^{-\theta_i x} - \chi(y \le x))^2 dx \\
&= \int_0^{y} (1 - \sum_{i=1}^{n} \lambda_i e^{-\theta_i x})^2 dx + \int_y^{\infty} (-\sum_{i=1}^{n} \lambda_i e^{-\theta_i x})^2 dx \\
&= \int_0^{y} 1 - 2\sum_{i=1}^{n} \lambda_i e^{-\theta_i x} + \sum_{i,j=1}^{n} \lambda_i \lambda_j e^{-(\theta_i + \theta_j)x} dx + \int_y^{\infty} \sum_{i,j=1}^{n} \lambda_i \lambda_j e^{-(\theta_i + \theta_j)x} dx \\
&= \int_0^{y} 1 - 2\sum_{i=1}^{n} \lambda_i e^{-\theta_i x} dx + \int_0^{\infty} \sum_{i,j=1}^{n} \lambda_i \lambda_j e^{-(\theta_i + \theta_j)x} dx \\
&= y + \sum_{i=1}^{n} \frac{2\lambda_i}{\theta_i}(e^{-\theta_i y} - 1) + \sum_{i,j=1}^{n} \frac{\lambda_i \lambda_j}{\theta_i + \theta_j}.
\end{aligned}$$

While it is possible to derive $\overline{\text{CRPS}}(G, H)$ for $G, H \in \mathcal{F}_{\text{exp}}$ analytically, we refrain from doing so to show how one might approach this problem in an applied setting. We use the estimate of $\overline{\text{CRPS}}(G, H)$ given by

$$\frac{1}{n} \sum_{i=1}^{n} \text{CRPS}(G, y_i), \tag{5.1}$$

where the y_i are independently sampled from the distribution H. We know that the map $y \mapsto \text{CRPS}(G, y)$ is $\mathcal{F}_{\text{exp}}$-integrable and thus the weak law of large numbers applies and

$$\frac{1}{n} \sum_{i=1}^{n} \text{CRPS}(G, y_i) \to \mathbb{E}_H[\text{CRPS}(G, Y)] = \overline{\text{CRPS}}(G, H)$$

in probability as $n \to \infty$.

5.3 Simulation

Our simulation study works as follows: We first define a "true" distribution G, i.e. the distribution we sample from. Then, we define some other distributions which may get predicted by forecasters. Finally, we generate one million independent realisations of $Y \sim G$ and use Formula 5.1 to estimate $\overline{\text{CRPS}}(G, H)$. We use the statistical software R.

The true distribution G we choose is

$$G = 0.2E_1 + 0.3E_3 + 0.5E_{10}.$$

This choice comes from the observation that the convex combination of exponential distributions with rates which differ by an order of magnitude cannot be easily approximated by a single exponential distribution. Furthermore, we take nice numbers for simplicity. See Polymenis [2025] for an example where a mixture of exponential distributions whose rates differ by an order of magnitude occurs and the demonstration why using the mixture is better. We see that in $\mathcal{F}_{\text{exp}}$, there is F with heavier tail than G, e.g. $F = E_{0.5}$. Furthermore, for $T = T_\rho$ the condition for statement ii) is met too as $T_\rho(F) = -0.5 > -1 = T_\rho(G)$, so for these choices of F, G and $T = T_\rho$, the assumptions of Theorem 4.4 hold.

So, for every $\varepsilon > 0$, there exists F_ε not tail equivalent to G such that $|\overline{S}(F_\varepsilon, G) - \overline{S}(G, G)| \leq \varepsilon$. In our simulation, we choose the same explicit construction as in the proof of Theorem 4.4, where we take $F_\lambda = \lambda F + (1 - \lambda)G$ and let λ be small enough.

Let us first have a look at the distributions we work with via their probability density function in Figure 5.1.

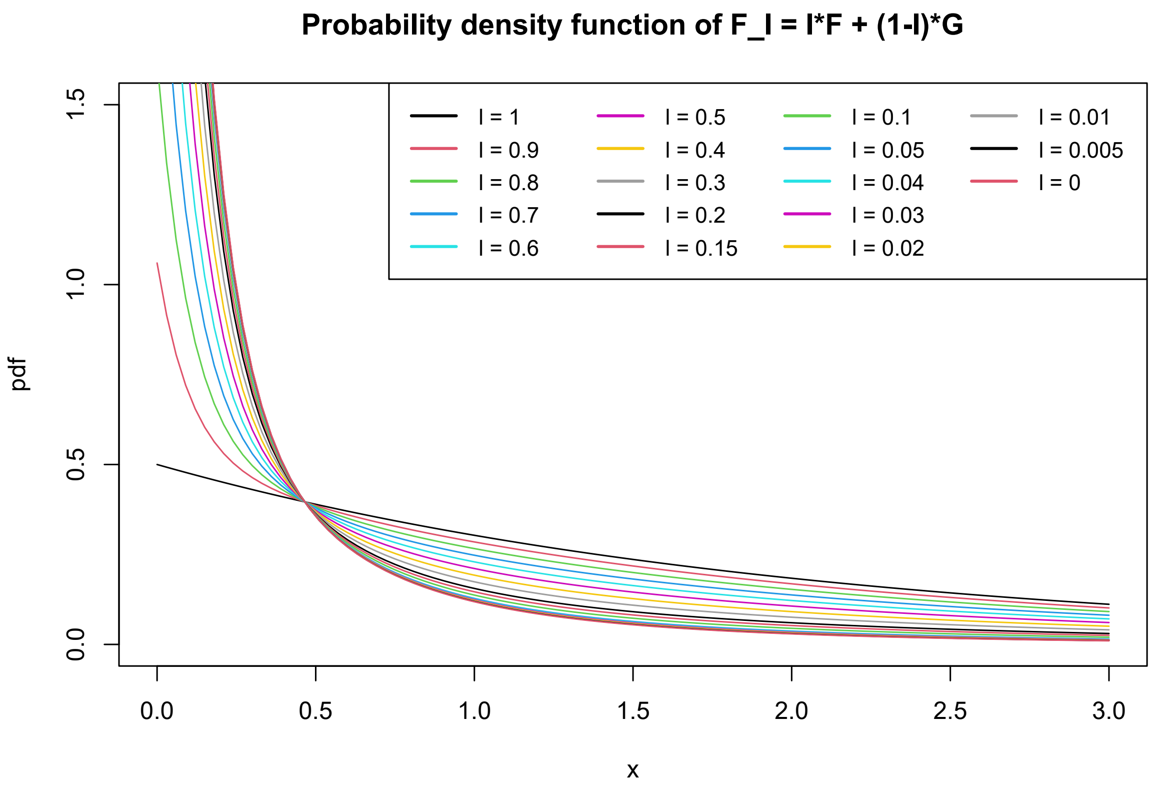


Figure 5.1: The probability density functions of various convex combinations of F and G. Although they vary greatly in initial value, all decrease more or less rapidly. Unsurprisingly, the pdf of F_λ is close to the one of G for small values of λ.

As the tails are harder to separate in Figure 5.1, let us look at the logarithm of the pdf to get a better feeling of the tail behaviour. The difference in tail behaviour becomes clearer in Figure 5.2.

Now let us look at the outcome of the simulation. Figure 5.3 and Figure 5.4 neatly show the predicted convergence by Theorem 4.4. Both are from the same run of the simulation with $n =$ 1,000,000 realisations to estimate $\overline{\text{CPRS}}(F_\lambda, G)$. We Note that for $\lambda = 0.005$, the estimate of $\overline{\text{CPRS}}(F_\lambda, G)$ equals 0.2312053, so it is only 0.008% higher than the estimate of $\overline{\text{CPRS}}(G, G)$ at 0.02311869.

In Figure 5.5 we can see that in a different run of the simulation with "only" 1,000 realisations we use for the approximation, the convergence does not seem to be monotone. This is can be explained by Remark 4.5, which says that due to sampling randomness and the fact that we have a finite sample, the average of the $\overline{\text{CRPS}}(F_\lambda, G)$ may be below $\overline{\text{CPRS}}(G, G)$ although $\overline{\text{CPRS}}(G, G)$

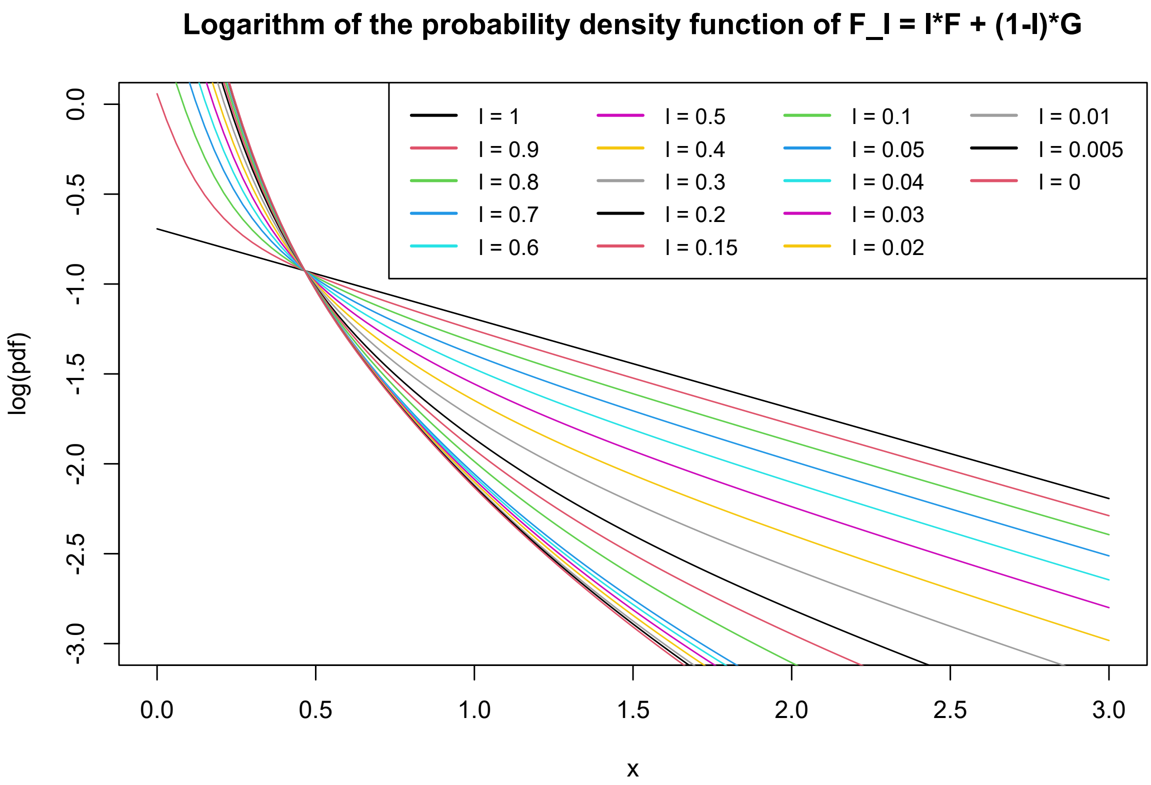


Figure 5.2: The logarithm of the probability density functions of various convex combinations of F and G. The difference in tail behaviour is clearer.

is the theoretical minimum. If the given distributions F_λ were the ones a decision maker had to evaluate, the decision maker would think that $F_{0.02}$ was closest to the real distribution as it had the lowest (approximated) expected score. As a consequence, the decision maker would have the wrong impression of the tail behaviour and could change decisions because of this.

From this finding, we see that the number of realisations we compare our distribution to matters a lot. We thus repeat the approximation of $\overline{\text{CRPS}}(F_\lambda, G)$ for different number of realisations of various orders of magnitude. We still consider the same distributions, i.e. values of λ as before. Figure 5.6 shows the fraction of times where the theoretical minimum is also the observed minimum for one thousand repetitions of each number of realisations being a power of ten between ten and one hundred thousand. For one hundred thousand realisations we find in approximately 97% of the cases the true minimum. This gives an empirical justification of letting our previous simulation run one million times.

5.4 Discussion

We find that choosing $\mathcal{F} = \mathcal{F}_{\text{exp}}$ for the considered distributions does have its applied justification by Polymenis [2025]. Then, if one is interested in the tail behaviour, using $T = T_\rho$ is a reasonable choice for this family. In this setting, the assumptions of Theorem 4.4 hold.

We indeed find the predicted convergence of $\overline{\text{CRPS}}(F_\lambda, G)$ to $\overline{\text{CRPS}}(G, G)$ as $\lambda \to 0$, even if

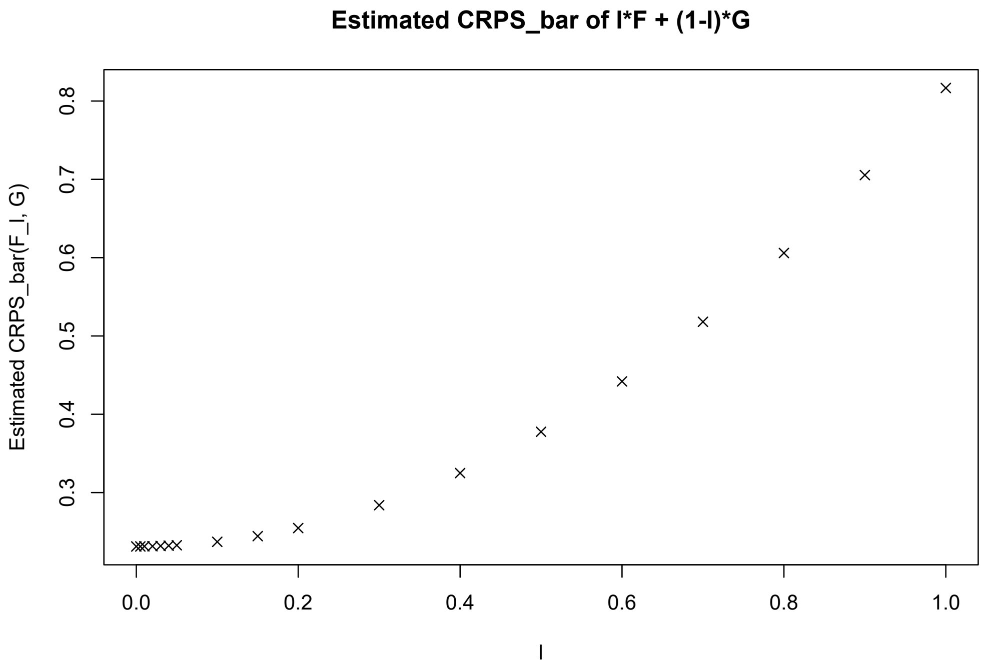


Figure 5.3: As the theory result predicts, the $\overline{\text{CPRS}}(F_\lambda, G)$ converges toward $\overline{\text{CPRS}}(G, G)$ as λ becomes smaller. This simulation ran with n =1,000,000 realisations to approximate $\overline{\text{CPRS}}(F_\lambda, G)$.

we only estimate $\overline{\text{CRPS}}(F_\lambda, G)$. Nevertheless, we see that for very small values of λ, we need many realisations for a good approximation of $\overline{\text{CRPS}}(F_\lambda, G)$ as we see that the lower bound of $\overline{\text{CRPS}}(G, G)$ can be violated even for $n = 10{,}000$ realisations in approximately 30% of simulation runs. In reality, one hardly ever has ten thousand let alone one million realisations of the random variable Y at hand and even if this is the case, for the convergence of our estimate to the true value to hold, the realisations need to be independent, which is a very strong assumption. This observation only makes the already bad news of Theorem 4.4 worse for practical applications.

Admittedly, the distributions $F_{0.005}$ and $F_{0.01}$ are in the list of compared distributions and distinguishing them from G is very hard. Future research could investigate how the minimal number of independent realisations for 95% confidence to choose the right minimum depends on the minimal λ used in the list of distributions considered. One could for example restrict all λ_i to be at least e.g. 0.01 or 0.05.

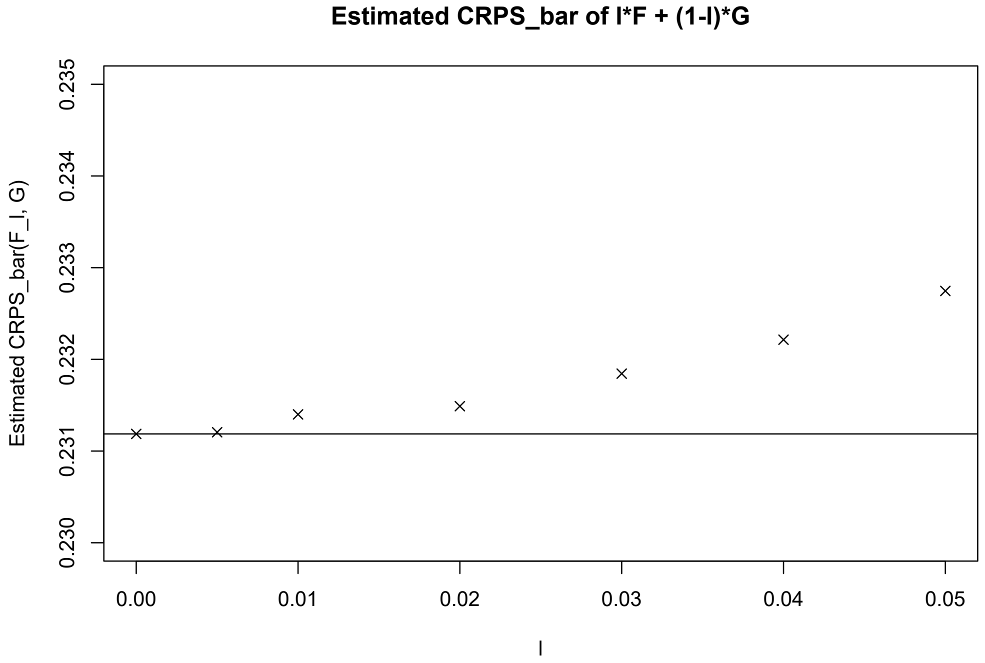


Figure 5.4: Close up of the simulated distributions with $\lambda \leq 0.05$. The vertical line represents the value of $\overline{\text{CPRS}}(G, G)$, i.e. when $\lambda = 0$. This simulation ran with n = 1,000,000 realisations to approximate $\overline{\text{CPRS}}(F_\lambda, G)$.

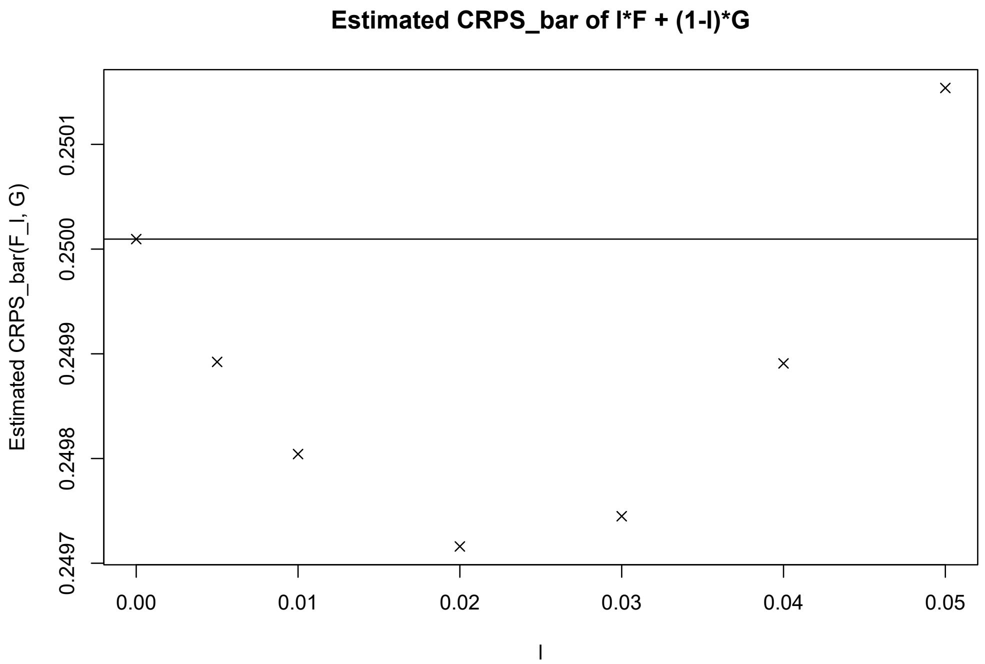


Figure 5.5: A second run of the simulation with “only” n = 1,000 realisations. The convergence does not seem to be monotone, which is possibly due to effects of randomness as remarked in Remark 4.5.

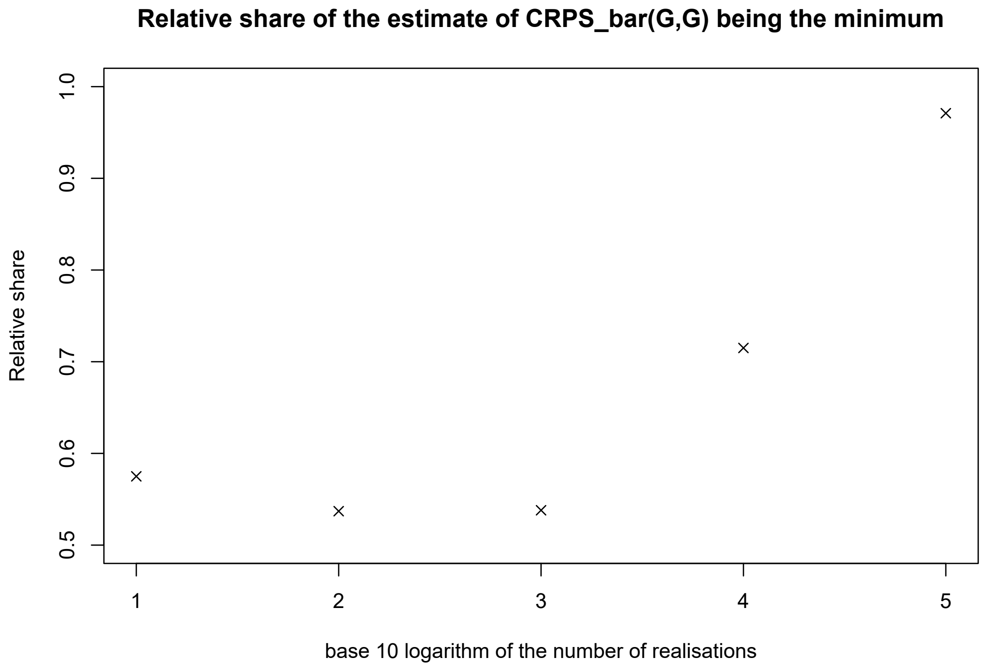


Figure 5.6: A comparison of the relative share of times where the theoretical minimum is the observed minimum for different numbers of realisations used for the approximation. Each number of realisations was repeated a total of $n_{\text{rep}} = 1{,}000$ times.

Chapter 6

Summary and Conclusion

In this thesis we started off with an introduction to scoring functions, which are a way to assess point forecasts. We introduced the concept of elicitability, had a look at some examples and counterexamples of elicitable functionals and motivated why it is desirable for a functional to be elicitable. It arose quite naturally to then define the elicitation complexity of a functional.

We went on to explore max-functionals, a special type of functional that arises when thinking about tail properties of distributions. Look at more max-functionals, which are rather made up for understanding than have practical applications, we nevertheless gained some insight on how max-functionals work. Future research might still be interested in further analysis of max-functionals to determine when they are a property of the tails and when not. Max-functionals turned out to have an infinite elicitation complexity with respect to two very broad classes and we have thus learned that the evaluation of their forecasts is very hard.

The idea to abandon point forecasts and look at whole distributions being forecasted turned out to not be the solution to our problems. Arbitrarily large differences in tail behaviour can still lead to small differences in expected score and thus remain undetected. We underlined these findings in a simulation study and identified even more problems which stem from the approximation of the expected score even under ideal conditions. Lastly, we had a look at how many realisations we need for our estimation to be good enough to reliably choose the correct distribution. This last point is certainly of big interest in practice but the finding that we need one hundred thousand independent realisations to choose the right distribution with a probability of 97% is bad from a practical point of view.

Future research could investigate how the minimal number of independent realisations for 95% confidence to choose the right minimum depends on the minimal λ used in the list of distributions considered. One could for example restrict all λ_i to be at least e.g. 0.01 or 0.05.

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[1] For further information please consult the ETH Zurich websites, e.g. https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html and https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html (subject to change).